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PALAEOMAGNETISM OF SILURIAN LAVAS OF SOMERSET AND GLOUCESTERSHIRE, ENGLAND

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Detailed alternating field demagnetisation of Upper Llandovery volcanics of the Mendip Hills and Gloucestershire has isolated remanence directions interpreted as primary from each of five sites. Well-defined high-coercivity secondary magnetisation is present in six samples of one site and low-coercivity secondary remanence is present in all samples from another site; the former component was apparently acquired in Permo-Triassic times. Primary directions of magnetisation show marked improvement in precision after correction for penecontemporaneous folding, and show a late Llandovery reversal in the sense R → N.

The group mean directions of magnetisation is $D = 243.5°$, $I = 47.5°$ (precision parameter $k = 29$). Petrographic examination confirms observations from magnetic properties that relict titanomagnetite (oxidation classes 3 to 5) is the remanence carrier in most samples. Hematite, probably mostly late magmatic in origin, is widely developed in all samples, but only the principal remanence carrier where it has thoroughly replaced the titanomagnetite. Low-coercivity remanence is apparently caused by weathering effects but there is no clear visible cause for secondary high-coercivity remanence carried by some samples.

The mean virtual geomagnetic pole position is close to Upper Silurian/Lower Devonian pole positions from other parts of Britain and defines a minimum apparent polar shift of 60° between late Ordovician and Upper Llandovery times. Reference to absolute age dates suggests that this shift took place between ca. 447 and 434 m.y. followed by slight polar movement between ca. 434 and 394 m.y.

1. Introduction

Palaeomagnetic studies of Lower Palaeozoic rocks [1] demonstrate that movement of the pole relative to Britain comprised a series of rapid shifts separated by longer periods during which the pole was practically stationary. This behaviour may have characterised polar movements during longer intervals of Precambrian and Phanerozoic time (e.g. [2] and [3]), but incomplete data coverage makes this less certain. Timings of Palaeozoic shifts are in need of closer definition because they are currently only based on secondary data from small amounts of material. This applies particularly to the shift under examination here. Ordovician rocks from Britain yield poles near 0°E, 10°S while Upper Silurian and Lower Devonian rocks yield poles in the vicinity of 325°E, 0°S [1]. Both of these groups of poles are now well established but the timing and magnitude of movement between them is not well defined because of the virtual absence of Silurian igneous rocks within Britain. Exceptions to this rule are exposures of volcanic rocks in Somersetshire and Gloucestershire, an investigation of which forms the subject of this note.

2. Geological outline of samples localities

The Silurian rocks of the Mendip Hills are exposed in a number of small inliers; they include about 200 m of pyroxene andesite flows and tuffs exposed by quarrying at Moon's Hill, Sunnyhill and Downhead. Geological information is summarised by Curtis [4]. The succession at the two first-named localities comprises andesite flows separated by a tuff overlying tuffs and shales containing probable Upper Llandovery fossils. Palaeomagnetic sites 2, 4 and 3 (stratigraphic order) were collected through

this sequence. These volcanics were tilted and partially eroded prior to eruption of another andesite overlain by sediments of Wenlockian age, and exposed at Downhead where site 1 was sampled.

The Silurian volcanics of Gloucestershire [5] comprise two lavas of similar petrology to the Somerset volcanics exposed near Tortworth (355.5E, 51.5N). They occur within, and are separated by, sandstones and shales of Upper Llandovery age, and succeeded conformably by Wenlock sediments. Palaeomagnetic results have been reported from 11 samples of these two lavas [6] but no systematic remanence was observed from the locality of site 5 of the present study in the lower lava (site T19 of ref [6]).

3. Palaeomagnetic results

Samples were cored using a motorised drill [7], oriented by sunsights and measured in the laboratory with a parastatic magnetometer. All samples were progressively demagnetised in increasing steps of 50 or 100 Oe until they ceased to show systematic or significant direction changes. Palaeomagnetic results for selected samples are shown in Fig. 1.

The ten samples from site 1 fall into two distinct groups with quite different magnetic properties. Four samples start with southerly declinations, show moderate to sharp falls in moment with progressive a.f. demagnetisation, and group about a mean $D = 49°$, $I = -46°$ ($k = 88$) after tilt correction and treatment in fields of 500–1000 Oe with about 5% of their initial moments remaining. The remaining six samples have comparable initial intensities of magnetisation but exhibit no fall in moment and slight change in direction on demagnetising to 1400 Oe. They group about a mean of $D = 52°$, $I = 42°$ ($k = 111$) quite different from that of other samples.

Both the high- and low-coercivity groups have comparable thermomagnetic characteristics and exhibit two stable Curie points, one between 70° and 160°C and the other between 520° and 540°C. There is little change in saturation magnetisation (J_s) of either the high- or low-temperature phases on heating, and since this could not be the case for a titanomagnetite with ulvöspinel in solid solution, the low Curie point probably represents a member of the hematite–ilmenite series. Demagnetisation characteristics of the high-coercivity group are commensurate with remanence residing in hematite [8]; this may not be clear from thermomagnetic characteristics because J_s for hematite is less than 0.5% than for magnetite [9] and its Curie point may not therefore be detectable.

All samples from site 2 show moderate falls in moment, and directions of magnetisation group about a well-defined mean of $D = 242°$, $I = 54°$. Thermomagnetic analyses give low J_s values with a single Curie point of 540°C. Site 3 samples exhibit slight fall in moments with no significant change in directions after treatment to 1400 Oe. They also give low J_s values with Curie points of approximately 550°C, and a suggestion of a second lower Curie point at 100–150°C. Both of these latter sites combine low J_s values with high coercivity suggesting a hematite–ilmenite member as a possible remanence carrier; for the reason given above thermomagnetic analyses might be ineffective in distinguishing this. All samples from site 4 show moderate to large falls in moment moving initially to cluster about a direction $D = 96°$, $I = 3°$ ($k = 40$) after loss of about 50% of their moments and demagnetisation in fields of 200–400 Oe, and then to a direction close to $D = 245°$, $I = 46°$ which is retained in fields of 1400 Oe with about 5% of the initial moments remaining. Two stable Curie points are present at 190°C and 510°C with no significant change in J_s after heating. By contrast behaviour of site 5 from Tortworth is simple: all samples show moderate to good stability grouping about a mean of $D = 239°$, $I = 62°$ which is retained after high-field treatment. J_s–T curves define a single Curie temperature of 540°C; this is stable against heating but J_s decreases by about 50%, interpreted as high-temperature oxidation of the stable magnetic phase to one with lower J_s, possibly oxidation of titanomagnetite to hematite.

The end result of the study of palaeomagnetic directions is that the two lower sites 2 and 4 are magnetised in the opposite direction to site 3 and the low-coercivity component of site 1. By reference to the apparent polar wander curve [1] this is seen to be a R → N transition with respect to the present Earth's field. Site 5 from Tortworth has the same polarity as the lower sites with which it may thus be comparable, and a reversal between the lower and upper lavas at Tortworth suggested by previous study [6] may be the same event as that recorded in the Mendip suc-

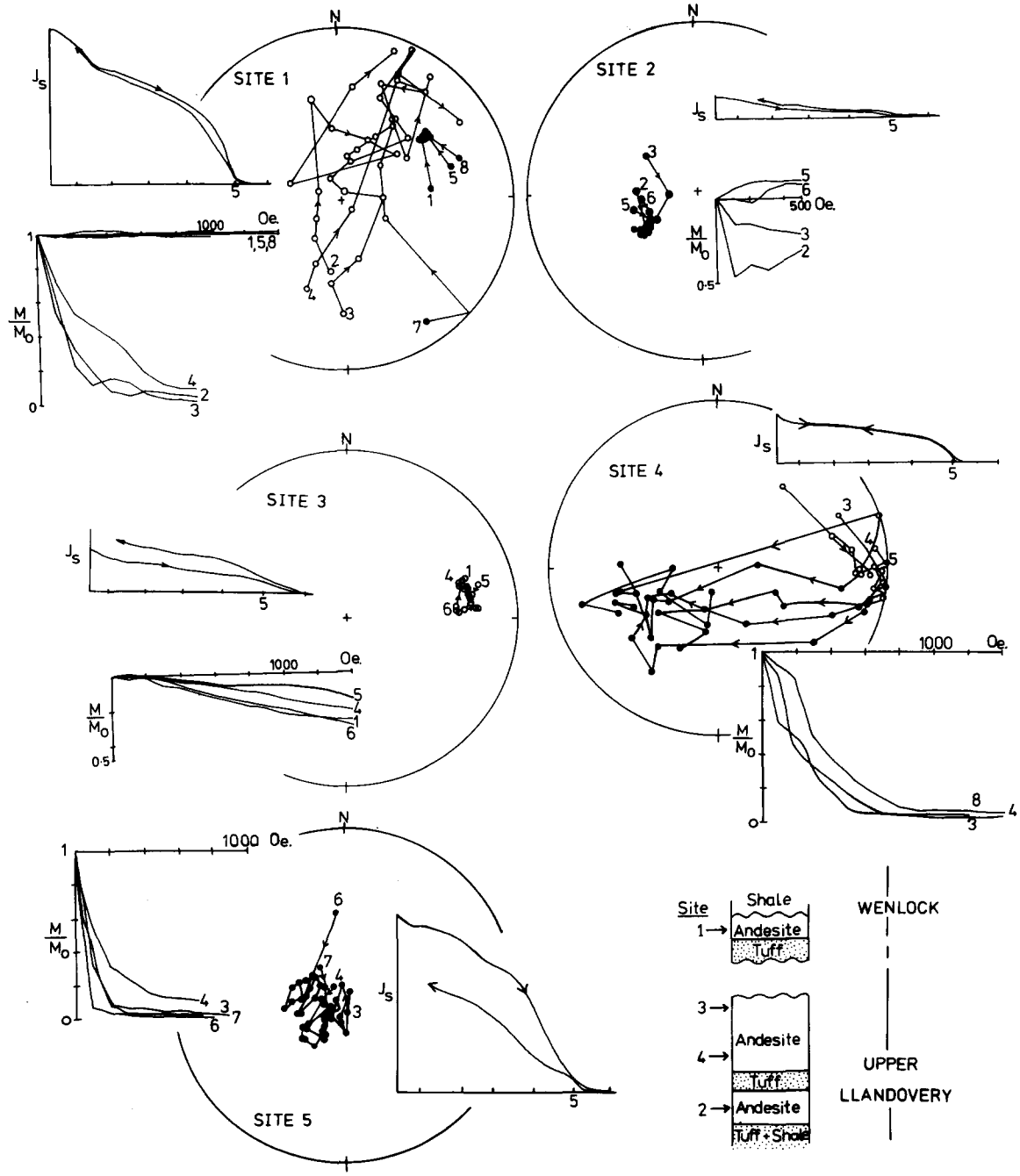


Fig. 1. Demagnetisation characteristics, J_s–T curves, and direction changes with demagnetisation in steps of 50 or 100 Oe for selected samples from Silurian lavas of Somerset and Gloucestershire. Divisions and numbers on the abscissa of the J_s–T curves represent temperature times 100°C. Projections are stereographic and closed symbols represent positive inclinations and open symbols represent negative inclinations. The lower right-hand diagram shows the approximate stratigraphic position of the sites from the Mendip Hills in the Silurian stratigraphic succession. All directions of magnetisation are dip corrected.

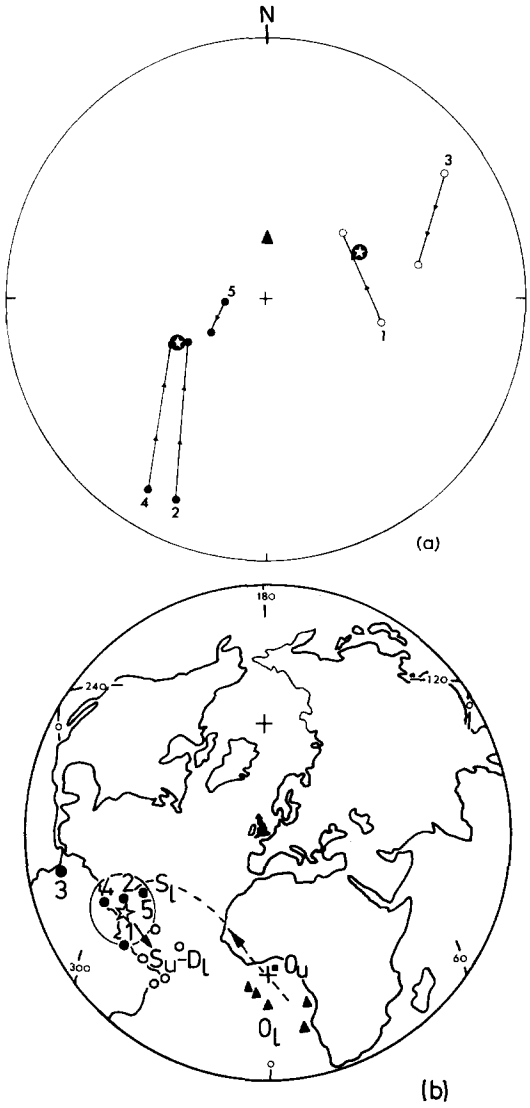


Fig. 2. (a) Site mean direction from the Silurian lavas of this study before and after correction for tectonic dip. The star is the group mean axis of magnetisation after dip correction and the large closed circle is the direction of the present geomagnetic field in the sampling region.

(b) Pole position and 95% confidence circle for this study in relation to Lower Ordovician (triangles), Upper Ordovician (square), Upper Silurian/Lower Devonian (open circles) poles. Poles for individual sites shown by closed circles.

cession (W.A. Morris, personal communication). These conclusions apply after correction for tectonic dip: sites 2 to 4 dip 65° in direction 350°E; site 1, 50° in a direction 160°E; and site 5, 15° in a direction 200°E. Dual polarity is recognisable only after dip correction and the grouping of sites with the same polarity is considerably improved Fig. 2a. Since tectonic dips were acquired shortly after eruption (partly prior to eruption of lava of site 1 in the case of sites 2–4), this test leaves no doubt that the direction of magnetisation derived from the five sites ($D = 243.5°$, $I = 47.5°$) is a true record of the geomagnetic field in Upper Llandovery time.

The dip-corrected direction of the higher coercivity component from site 1 is not interpretable in the context of Phanerozoic geomagnetic field directions from Britain; however, the direction uncorrected for dip ($D = 27°$, $I = 15°$) might be Carboniferous in age [10] but the low coercivity component of site 4, even when dip corrected ($D = 90°$, $I = 13°$), lies close to no known post-Devonian field direction for Britain, and its origin is not clear. The explanation of the site 1 high-coercivity component is reasonable from a geological standpoint since the region was exposed to extensive weathering and erosion following the Hercynian orogeny [11,12], but the occurrence of two quite different magnetisations in different samples of one outcrop is noteworthy and merits petrologic study.

4. Origin of NRM's

All samples examined petrologically (with Reichert Zeto-pan Pol microscope to X 1900 magnification) exhibit varying degrees of hematisation; hematite is developed as small grains in the groundmass, fine veinlets, and as rims replacing primary titanomagnetites. Three petrological observations readily account for magnetic properties. Firstly, site 2 has titanomagnetites almost entirely consumed by secondary aggregates of titanohematite and pseudobrookite; secondly, site 3 has grain sizes an order finer than the other two sites and conversion to hematite is almost total; and thirdly, site 4 is more weathered than the remainder with extensive amount of iron hydroxides (possibly geothite as indicated by red internal reflections) in the groundmass. The first two observations account for the distinctive magnetic characteristics of sites 2 and 3, and the third observation gives a possible reason for the soft secondary component removed by alternating field cleaning in site 4.

Two important questions are posed by ubiquitous

TABLE 1

Mean directions of palaeomagnetic sites in Silurian lavas of Somerset and Gloucestershire, after a.f. cleaning and tilt correction

Site No.	No. of samples	D	I	R	a_{95}	k	Palaeomagnetic pole	
							°N	°E
1	4	48.5	−46.2	3.96	10.0	88	1	134
2	6	241.9	+53.7	5.96	6.1	120	11	308
3	7	77.3	−26.8	6.98	3.6	277	−3	104
4	8	245.1	+46.1	7.59	13.7	17	7	301
5	9	239.1	+62.2	8.78	8.6	37	18	315
Mean of directions and pole positions with sites 1 and 3 reversed	5	243.5	+47.5	4.88	13.6	29	8.1	308.7

Note: D = declination and I = inclination of remanence vector, R = length of resultant of samples, a_{95} = semi angle of cone of 95% confidence for mean direction, $k = (N-1)/(N-R)$ = estimate of Fisher's κ parameter where N is number of samples.

presence of hematite: they relate to its paragenesis and the extent to which it contributes to the NRM. Titanomagnetite is widely developed in sites 1, 4 and 5 subdivided by ilmenite lamellae variable altered to secondary aggregates (oxidation classes 3 to 5 [13]). Although hematite is in most cases more abundant than titanomagnetite, it is the latter mineral which dominates magnetic remanence until it is all but destroyed (as in sites 2 and 3). A late magmatic origin for the hematite is favoured since primary hematite is crystallised in the groundmass and thoroughly replaces margins of all titanomagnetite grains present. Petrographic examination reinforces a point made by Storetvedt [14] that it is not always possible to identify remanence carriers by petrologic examination alone, and this is particularly emphasised by thorough examination of site 1 which reveals no clear petrographic difference between high- and low-coercivity samples. Both contain abundant titanomagnetite and high-coercivity samples sometimes contain *less* hematite; the only difference seems to be in degree of incipient alteration of titanomagnetites, and it may be that sporadic submicroscopic alteration has effectively destroyed primary titanomagnetite-held remanence.

5. Virtual geomagnetic pole

The derived virtual geomagnetic pole position from this study (308.7°E, 8.1°N) provides a Lower Silurian pole from Britain close to Upper Silurian and Lower Devonian poles [1]. It is removed by approximately 60° from typical Ordovician poles. A single Upper Ordovician pole [15] is supplemented by further data (author's unpublished results) which demonstrate that the pole remained in this general region until late Ordovician time. This shift is thus dated as Lower/Middle Llandovery in age and on the absolute time scale it postdates the age of Caradocian Bentonites [16] of 447±5 m.y. and predates an Upper Llandovery age of 434±6 m.y. [17] closely confirmed by an additional Llandovery age [18]. Dated rocks giving established poles in the typical Upper Silurian/Lower Devonian group range from ca. 415 m.y. to 394±5 m.y. in age, and are displaced by approximately 15° from the Lower Silurian pole given here (Fig. 2b).

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